Rehbar Khan

me25mtech11015@iith.ac.in | rahbarkhan100@gmail.com | +91 9634705626 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.65
Sayyid Hamid senior secondary school	Class XII (AMU Board of secondary and senior secondary examination)	2020	85.00%
International Indian School Dammam	Class X (CBSE)	2018	91.40%

PROJECTS

Autonomous USV-UAV Maritime Inspection System | *ROS 2, Isaac Sim, PX4, Gazebo, OpenCV*

- Built a two-vehicle system where a wave-stabilised USV launches and recovers a UAV, integrating a from-scratch nonlinear 6-DOF Fossen vessel model (mesh-derived hydrostatics, JONSWAP waves, wind) with UKF state estimation and a radar-based Kalman-tracking + DWA avoidance stack.
- Implemented the aerial mission layer over **unmodified PX4 firmware** via ROS 2 / uXRCE-DDS -- deck-motion launch gating with hysteresis, and an autonomous inspection-and-recovery mission -- keeping the flight controller a black box so the stack transfers to hardware unchanged.
- Delivered vision-guided deck landing (ArUco + downward camera ->OpenCV pose ->PX4 vision_target_estimator ->precision landing), **cutting landing error from 36 cm to 9 cm (74 %)**; tuned PX4's search/handover altitudes and filter covariances from a measured 1-6 m detection sweep. Validated with 60+ automated tests.

Deep RL Motorcycle Control | PyTorch, MuJoCo, PPO, ROS2, C++, ONNX | github.com/Rehbar25/moto-rl

- Built a MuJoCo motorcycle model (rake and trail steering geometry, validated against steady-turn physics) and trained **PPO** controllers for balance and road following, learning **steering and throttle from scratch** with 96-100% success on random roads.
- Trained a **domain-randomized policy** (mass, center of mass, tire grip, actuator strength, 0-60 ms delay, sensor noise) that kept **90% success under combined worst-case disturbances**, vs. **17%** for a tuned cascaded classical controller and **13%** for a non-randomized policy.
- Validated **sim-to-sim transfer** by rebuilding the vehicle in **PyBullet**: the randomized policy kept **97%** success in the unseen engine, vs. **67%** without randomization.
- Deployed the policy as a **C++ ROS2 (Humble) node with ONNX Runtime**: **10 µs inference** and a **0.19 ms** end-to-end control loop at a steady 50 Hz, with closed-loop behavior identical to PyTorch. Added pytest smoke tests and GitHub Actions CI.

ERPNext Implementation Simulation - Drone Manufacturing | *ERPNext, Python, REST/SOAP APIs, SQL, Docker*

- Delivered a full-lifecycle ERP implementation for a simulated drone manufacturer: requirements and fit-gap analysis (13 requirements), configuration of **Inventory, Procurement, Order Management, Manufacturing (multi-level BOMs, MRP) and Costing**, UAT and a cutover plan with rollback.
- Built a Python **data-migration pipeline** that cleansed legacy Excel data (**81 data-quality issues**) and loaded it via the ERPNext REST API with **19/19 reconciliation checks** passed; ran end-to-end **Procure-to-Pay and Order-to-Cash** cycles where MRP cut purchasing to **2 of 23 components**.
- Developed a **SOAP supplier price-sync** with approval controls and an **idempotent REST order-intake API**, validated by **60 automated tests**; wrote client-facing **status reports, issue logs and user guides** in plain language.

POSITIONS OF RESPONSIBILITY

Placement Coordinator, Office of Career Services, IIT Hyderabad

- Served as Placement Coordinator under the Office of Career Services, IIT Hyderabad, coordinating placement drives, liaising with recruiters, and facilitating communication between students and companies.

Finance Coordinator, Prayas IIT Hyderabad

- Served as Finance Coordinator for Prayas, IIT Hyderabad, managing budgeting, fund allocation, and financial record-keeping for the organization's initiatives.

Teaching Assistant, IIT Hyderabad

- Serving as a Teaching Assistant at IIT Hyderabad while pursuing an M.Tech in Robotics and Intelligent Systems, assisting in course delivery, conducting tutorials/lab sessions, and supporting evaluation for courses related to robotics and autonomous systems.

SKILLS

Technical Skills: Isaac Sim, Gazebo, PX4/ArduPilot, MATLAB Simulink, SolidWorks, Fusion, Solid Edge, ABAQUS, Ansys, LTSpice, Bambu Studio, MuJoCo.

Programming skills: C/C++, Python, MATLAB, ROS/ROS2, OpenCV, PyTorch, Git, Linux/Bash.

RELEVANT COURSES

Theory Courses: Control Systems Design, Probability and Optimization, Machine Learning, Soft Robotics, Autonomous Robotic Systems, Marine Robotics,

Lab Courses: Sensors and Actuators, Computer Vision, Robotics and Automation Lab, Advanced Robotics Lab.

PUBLICATIONS

[1] Sinha Roy, A., Khan, R., Anand, V. K., Kumar, P. R., and Pandey, A. K., "Hierarchical Control of Cooperative Quadrotor Payload Transport," Submitted to ASME IMECE 2026.

[2] R. Khan, I. Rasool, M. Afzal, and A. A. Khan, "Powder Bed Fusion Techniques in Metal 3D Printing: A Review," AMM, vol. 922, pp. 67-75, Aug. 2024, doi: 10.4028/p-ny5hlx.

EXTRACURRICULAR

Coordinated the 10-day **Autonomous Robotics Summer Immersion & Simulation Grand Prix 2026**, overseeing technical sessions, participant engagement, and event execution.